JACOB (JAKE) CARLSON

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HARVARD UNIVERSITY

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Education

Harvard University
Ph.D. Economics, 2022 to 2027 (expected)

Yale University
B.A. Statistics and Data Science, 2020
B.A. Ethics, Politics, and Economics, 2020
Summa cum laude, distinction in both majors

Fields

Primary: Econometrics
Secondary: Machine Learning Methods for Economics

References

Isaiah Andrews
iandrews@mit.edu

Melissa Dell
melissadell@fas.harvard.edu

Neil Shephard
shephard@fas.harvard.edu

Davide Viviano
dviviano@fas.harvard.edu

Fellowships & Awards

Rising Stars in Data Science, University of Chicago, 2026
PhD Fellowship Finalist, Two Sigma, 2026
Presidential Scholarship, Harvard University, 2022
Undergraduate Thesis Prize Winner, Forethought Foundation for Global Priorities Research, 2020
Phi Beta Kappa, Yale University, 2020

Teaching

Causal Inference in Economics (PhD, Spring 2025, 2026, Davide Viviano)
Time Series (PhD, Fall 2024, 2025, Neil Shephard)
Unleashing Novel Data at Scale (PhD, Spring 2021, Melissa Dell)

Employment

Harvard University, Data Science Fellow in Deep Learning, 2020 to 2022

Job Market Paper

Interpretable Discovery from Unstructured Data: A High-Dimensional Approach

We propose an automatic, general-purpose framework for making discoveries from unstructured data (e.g., text data from open-ended surveys of economic beliefs). The framework leverages recent methods from the literature on AI interpretability to transform unstructured datasets into high-dimensional, structured datasets of interpretable concept measurements; specifies concept-level parameters and null hypotheses based on this transformed dataset; tests these hypotheses using algorithms validated by new results in high-dimensional multiple testing, producing a selected set ("discoveries"); and both generates and evaluates human-interpretable natural language descriptions of these discoveries. The proposed framework has few researcher degrees of freedom, is robust to data snooping, mitigates under-exploration, and facilitates fast and inexpensive sensitivity analysis and replication. We revisit applications to recent descriptive and causal analyses of unstructured data in empirical economics, and find this framework is able to automatically replicate existing discoveries, add nuance to others, and make entirely new discoveries as well.

Conference Publications

Carlson, Jacob, Tom Bryan, and Melissa Dell. "Efficient OCR for Building a Diverse Digital History." *ACL* (2024). **Oral presentation.**

Dell, Melissa, Jacob Carlson, Tom Bryan, Emily Silcock, Abhishek Arora, Zejiang Shen, Luca D'Amico-Wong, Quan Le, Pablo Querubin, Leander Heldring. "American Stories: A Large-Scale Structured Text Dataset of Historical U.S. Newspapers." *NeurIPS* Datasets and Benchmarks Track (2023).

Bryan, Tom, Jacob Carlson, Abhishek Arora, Melissa Dell. "EfficientOCR: An Extensible, Open-Source Package for Efficiently Digitizing World Knowledge." *EMNLP* System Demonstrations (2023).

Shen, Zejiang, Ruochen Zhang, Melissa Dell, Benjamin Lee, Jacob Carlson, and Weining Li. "LayoutParser: A Unified Toolkit for Deep Learning Based Document Image Analysis." *ICDAR* (2021). **Oral presentation.**

Journal Publications

Andrews, Isaiah, Harvey Barnhard, and Jacob Carlson. "True and Pseudo-True Parameters." Forthcoming, *Econometric Theory*.

Carlson, Jacob, Trevor Incerti, and P.M. Aronow. "Dyadic Clustering in International Relations." *Political Analysis* (2024).

Working Papers

Carlson, Jacob and Neil Shephard. "When are time series predictions causal? The potential system and dynamic causal effects." Working paper (2026).

Carlson, Jacob and Melissa Dell. "A Unifying Framework for Robust and Efficient Inference with Unstructured Data." Working paper (2025).

Seminars & Conferences

2026: OpenAI AGI Economics Conference (OpenAI, September); ESIF Economics and AI+ML Meeting (Cornell, June); AI and Economics Summer Conference (Chicago Booth, August)
2022: ASSA Annual Meeting (Virtual, January)

Referee Service

Journal of Political Economy

Languages

English (Native)

Personal Information

Citizenship: U.S.A.